# Predicting Global Carbon Emissions by Utilizing the Energy Mix

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**Abstract**

***Background*:** A recent United Nations report indicates that the global surface temperature has been rising by nearly 0.2°C per decade since 1970 and has accelerated over the last 15 years, with projections to reach the 1.5°C warming limit by 2030. This trend is largely attributed to the high concentration of carbon dioxide ($CO_2$), which is affecting the environment and causing increased forest fires and glacial melts. The aim of this study is to delve into the crux of this issue and assess the impact of $CO_2$ emissions resulting from energy consumption and the energy mix.

***Methods*:** Utilizing the most recent publicly available data from 1983 to 2023, a flexible regression model is developed to examine the association between global primary energy consumption and $CO_2$ emissions over the last 40 years. The model is applied to project $CO_2$ emissions based on current trends and mitigation strategies by increasing renewable and nuclear energy in the energy mix.

***Results*:** The analysis shows that $CO_2$ emissions are significantly influenced by the energy mix, including fossil fuels, renewable sources, and nuclear energy. While both global emissions and energy consumption are projected to rise, increasing the share of renewable and nuclear energy up to 50% and 70%, respectively, could lead to a reduction in $CO_2$ emissions by one-third and two-thirds by 2030.

***Conclusions*:** This study demonstrates that with enhanced global control, regulations, and environmental standards, we can curb the growth of $CO_2$ emissions by transitioning to more renewable and sustainable energy sources.

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# Predicting Global Carbon Emissions by Utilizing the Energy Mix

## 1. Introduction

A recent United Nations report indicates that the global surface temperature has been rising by nearly 0.2°C per decade since 1970 and the increase has accelerated over the last 15 years, with projections to reach the 1.5°C global warming limit by 2030 (Hansen et al., 2023; Hausfather, 2024). This trend is largely attributed to high concentrations of carbon dioxide ($CO_2$) and greenhouse gas emissions, which are beginning to affect the environment by causing coral reef bleaching, increased forest fires, and glacial melts.

In the past few decades, the increase in global power generation has typically come from three different types of sources: fossil fuels (coal, oil, natural gas), renewable energy (solar, wind, hydro, biomass), and nuclear energy. Every year, gigatonnes of CO2 and greenhouse gases are emitted into the atmosphere, with fossil fuels as a main contributor. The massive increase in power consumption is attributed to many fast-growing industrial sectors, such as transportation, astronomy, information technology, high-performance computing (HPC), and generative artificial intelligence (GenAI) applications. For example, according to Goldman Sachs research reports, the power demand of HPC data centers was relatively stable at about 200 terawatt-hours per year before 2020, even though the workload tripled. This was largely due to increased efficiency in data centers and improved utilization rates of various energy sources. However, due to the revolution of AI, data centers' power demand is expected to outpace efficiency improvements, with a 160% increase in power consumptions by 2030 (Goldman Sachs Report., 2024). Without significant intervention, CO2 emissions will continue to rise at an accelerated rate with the increasing demand for power consumption (Andrae & Edler, 2015; Freitag et al., 2021; International Energy Agency, 2024).

The aim of this study is to delve into the crux of this issue and assess the impact of $CO_2$ emissions resulting from energy consumption and the energy mix. I sought to develop a flexible model to examine the association between global primary energy consumption and $CO_2$ emissions over the last 40 years using the most recent publicly available data from 1983 to 2023. The model is then applied to project $CO_2$ emissions in 2030 based on current trends and to explore impact of certain mitigation strategies, such as increasing renewable and nuclear energy in the energy mix.

## 2. Methods

Through an extensive review of published papers and an online search of publicly available datasets, the following datasets were identified to develop models for studying our research problem.

2.1 Carbon Intensity by Energy Sources for Electricity Generation

Electricity generation causes greenhouse gas emissions; however, the amount of greenhouse gases produced varies significantly depending on the energy sources used. Public data from Our World in Data (https://ourworldindata.org/safest-sources-of-energy) and the IPCC — Intergovernmental Panel on Climate Change report were used. Table 1 and Figure 1 list and compare the carbon intensity accounted for their lifecycle greenhouse gas emissions (in units of $gCO_2e/KWh$), indicating the amount of carbon dioxide equivalents (CO2e) emitted per kilowatt-hour (kWh) of electricity consumed.

**Table 1.** Carbon Intensity by Electricity Supply Technologies

| Energy Type | Coal | Oil | Solar | Natural Gas | Hydropower | Biomass | Wind | Nuclear energy |
|---|---|---|---|---|---|---|---|---|
| Cabon Intensity | 910 | 720 | 48 | 440 | 24 | 230 | 12 | 12 |

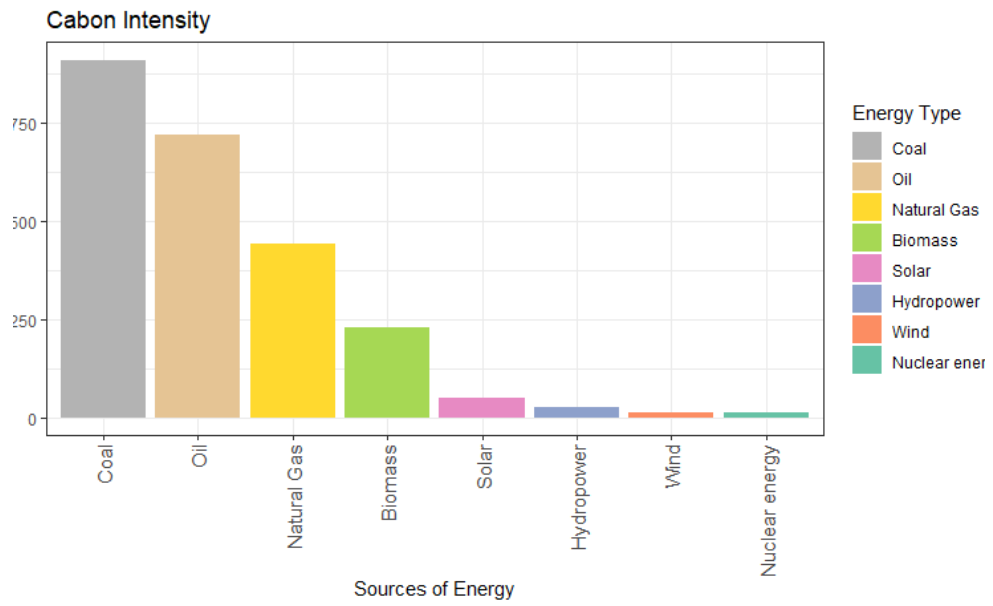


**Figure 1.** Carbon Intensities of eight energy sources.

2.2 Global Primary Energy Consumption by Source

Global energy mix data from Our World in Data (https://ourworldindata.org/energy-mix) is available for open access. For each of the eight energy sources in Table 1, the energy consumption in terawatt-hour (TWh) worldwide was extracted from 1983 to 2023.

2.3 Global Annual $CO_2$ Emissions

Global $CO_2$ emissions data from Our World in Data (https://ourworldindata.org/grapher/annual-co2-emissions-per-country) is available for open access. The $CO_2$ emissions data (in the units of GtCO2e = Billions Tonnes of carbon dioxide equivalent) was extracted from 1983 to 2023.

2.4 Statistical Analysis

I took several steps to develop models to assess the total $CO_2$ emissions as a function of energy consumption from 1983 through 2023. First, I proposed and computed a new quantity, called the worldwide Energy Mix Index (EMI), to quantify the contribution of $CO_2$ emissions by

the annual energy mix. Then, to account for the relationship between $CO_2$ emissions and the energy mix that may change over time due to the recent advances in technologies, I applied a spline-based varying-coefficient regression model to estimate global $CO_2$ emissions resulting from the energy mix index. Additionally, to explore the future trends until 2030, I predicted the changes of the energy mix index from 2024 to 2030 based on trends for each of the eight primary energy sources, then applied my varying-coefficient regression model to predict the future global $CO_2$ emissions. To assess the impact of different energy sources and mixes, I explored two possible scenarios with increased proportions of renewable energy and nuclear energy. The least squares method was used to estimate the coefficients of all regression models. The mathematical modeling and analysis were performed using the statistical Computing programming language R version 4.2.2 (https://www.r-project.org/).

## 3. Results

### 3.1 The Energy Mix Index (EMI)

As shown in Table 1, the amount of greenhouse gases produced varies significantly depending on the sources of energy. Therefore, the same energy consumption with different energy mixes, i.e., components of the energy sources, will produce different amounts of $CO_2$ emissions. I propose to quantify the $CO_2$ emission contributions over time as the weighted sum of energy consumption from various sources, with the corresponding carbon intensities as weights:

$$EMI(t) = \sum_{i=1}^{n} E_i CI_i$$

where $E_i$ and $CI_i$ are the amounts of energy consumption and carbon intensity for energy source $i$, $i$=1,…, n. Based on data in Sections 2.1-2.2, EMI ($t$) was computed in 1983–2023 (Figure 2).

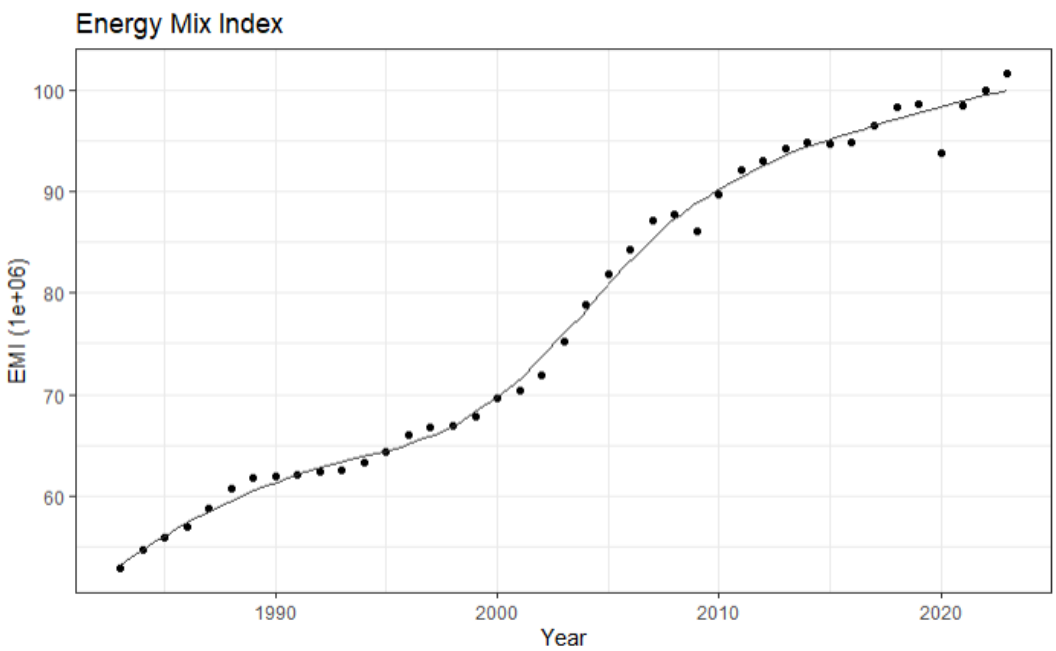


**Figure 2.** The Energy Mix Index based on the global primary energy consumption by 8 sources in Sections 2.1-2.2. A smoothing line is added to the annual estimate, indicating that the rate of change varied over time with several change-points.

Although there were increasing demands for solar, wind, and hydro power sources, the energy mix was still dominated by coal, oil and natural gas, which accounted for over 75% of total energy generation. This resulted in the increasing trend of the EMI over the past decades.

3.2 Global $CO_2$ Emissions Model based on the Energy Mix Index

To model the association between global $CO_2$ emissions and the energy mix index, a spline-based varying-coefficient regression model was applied to allow their relationship to change over the 40-year span (Hastie et al., 2008). Let a) the global $CO_2$ emissions, $Y(t)$, be the response variable, b) the energy mix index, EMI(t), be the predictor accounting for the different proportions of various energy sources in the annual global energy consumption, and c) $\varepsilon$ be the error term, the varying-coefficient regression model can be written as follows,

$$Y(t) = \beta_0 + \beta_1(t)EMI(t) + \varepsilon \qquad (1)$$

Where $\beta_1(t) = \gamma_0 + \gamma_1 t + \gamma_2 t^2 + \gamma_3(t - t_1)_+^2 + \gamma_4(t - t_2)_+^2$ was the coefficient curve based on quadratic splines with two change-points at 1999 and 2010 (Figure 3).

Unlike the standard regression model with fixed coefficients over time and no change-points, this varying-coefficient model (1) was more flexible and could be used to predict global CO2 emissions as a function of energy consumption. Moreover, the coefficient curve was positive and increasing (Figure 3A), suggesting that $CO_2$ emissions are increasing with energy consumption but the rate of change has been more rapid in the recent decades compared to the earlier decades. Figure 3B shows that the predicted values closely aligned with the observed ones, indicating that the model provided a good fit.

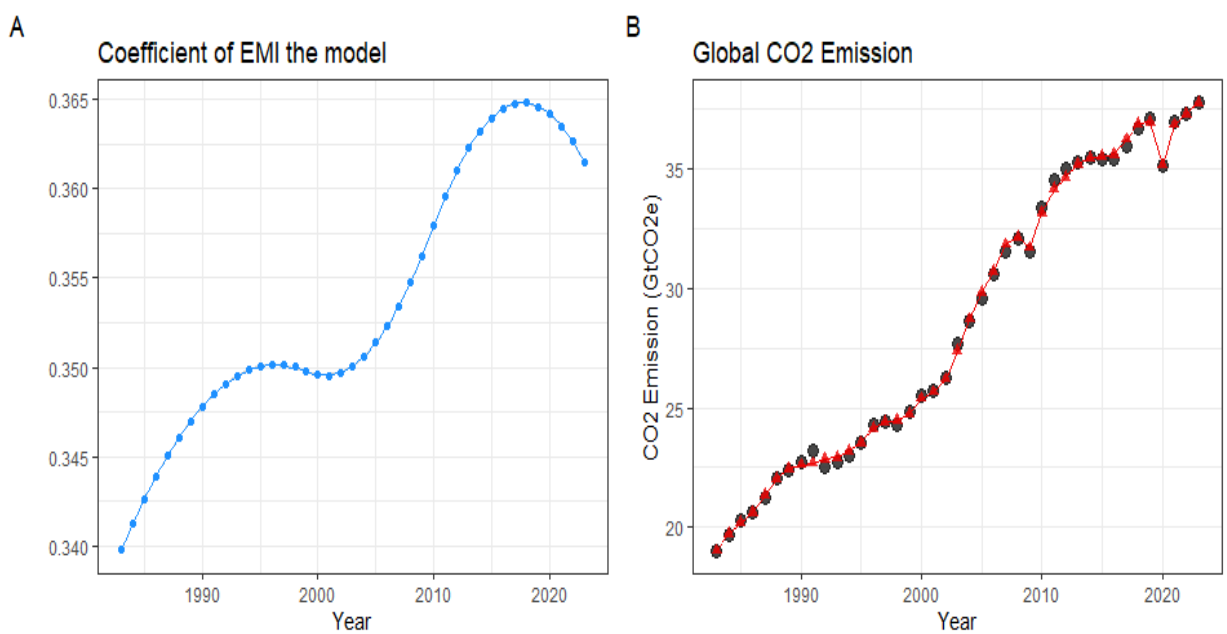


**Figure 3**. Modeling global CO2 emissions based on the Energy Mix Index. (A) Coefficient of the energy mix index, positive and varying over time; (B) Observed $CO_2$ emissions (black dots) and predicted values (red triangles) based on the above varying-coefficient regression model (1).

3.3 Predicting Future Trends of Energy Consumption and CO2 Emissions

Using the data in Section 2.2, I applied a linear spline model with two change-points at 1999 and 2010 to predict future power consumptions for eight energy sources (Figure 4).

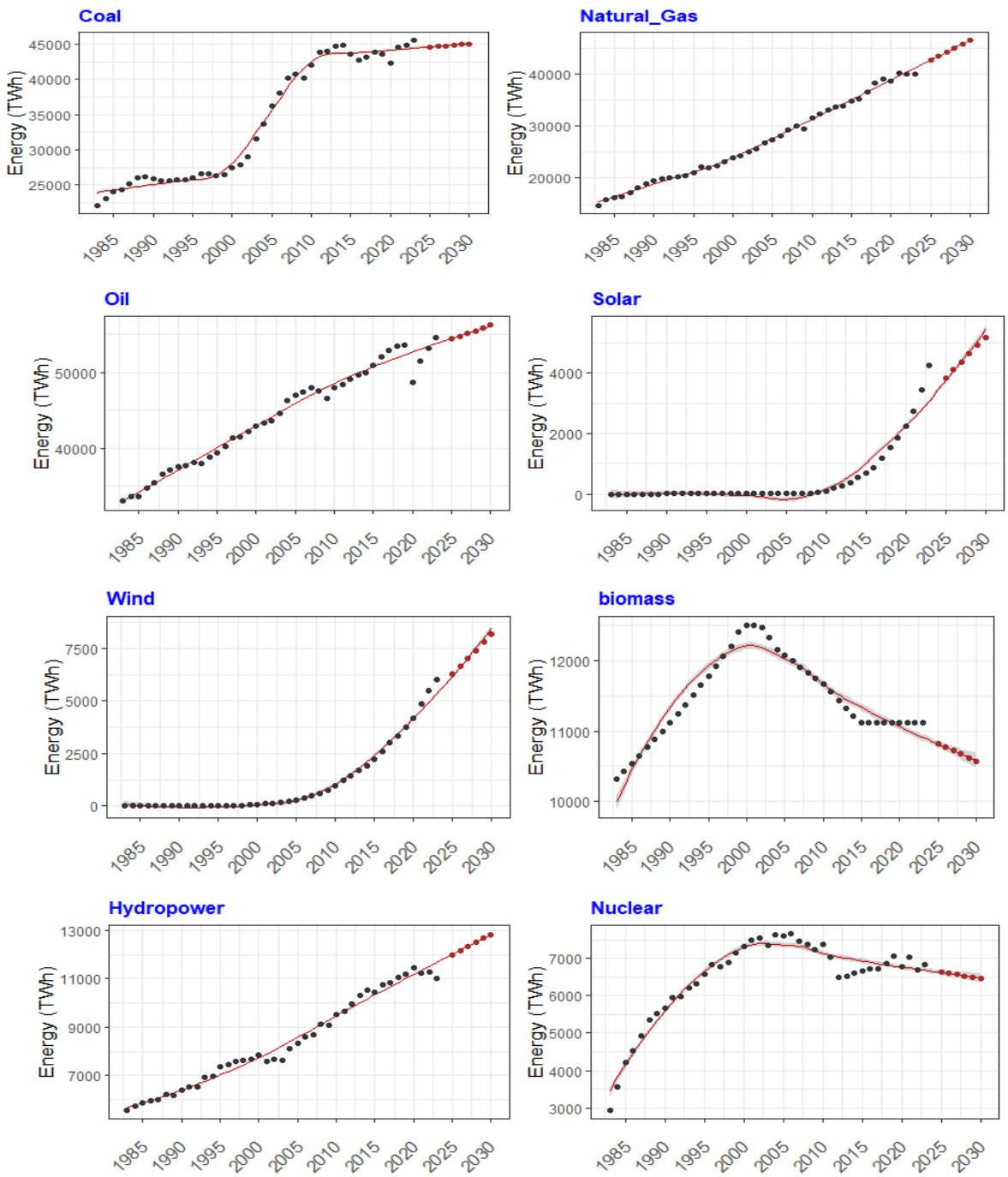


**Figure 4**. Projected power consumption by eight sources of primary energy. In each plot, the projected values are indicated with red dots and the fitted trend is added with a red line.

Therefore, the energy mix index, EMI(t), can be calculated from $t$ = 2024 to 2030, then plugged in the varying-coefficient regression equation (1) to predict the $CO_2$ emissions from 2024 to 2030 (Figure 5). Based on these prediction results, if the global energy consumption pattern remains similar, the $CO_2$ emissions are projected to continue to rise until the end of the decade (Table 2).

**Table 2**. Predicted $CO_2$ Emissions (GtCO2e).

| Year | 2024 | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 |
|---|---|---|---|---|---|---|---|
| CO2 emissions (GtCO2e) | 37.96 | 38.19 | 38.41 | 38.61 | 38.80 | 38.98 | 39.15 |

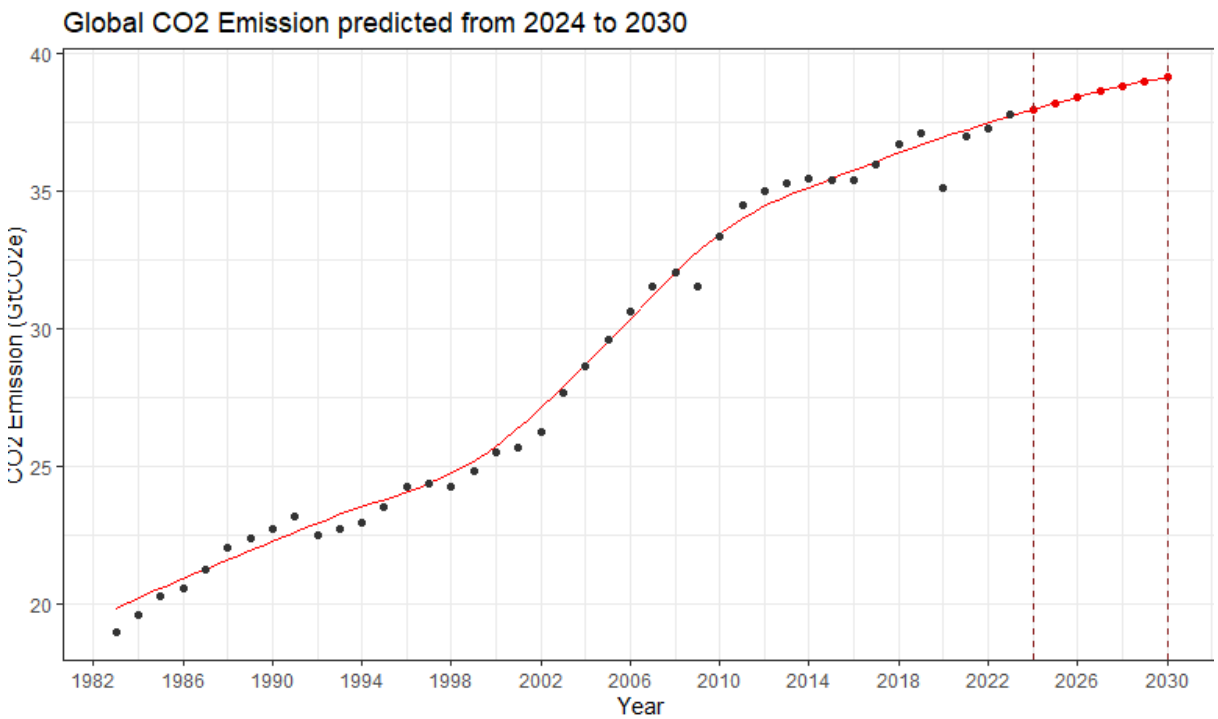


**Figure 5.** The estimated trend in global $CO_2$ emissions. The red line is the fitted line based on observed energy consumption and energy mix from 1983-2030. The red dots represent the predicted $CO_2$ emissions from 2024 to 2030 based on the regression model (1).

3.4 Predicting Trends of CO2 Emissions in 2030 Using Different Energy Mixes

As of 2023, the estimated global primary energy mix included oil at 30%, coal at 25%, natural gas at 22%, biomass at 6%, solar at 2.3%, wind at 3.3%, hydropower at 6%, and nuclear at 3.7%. Thus, the total proportion of renewable energy is less than 20% and nuclear energy is less than 5%. Using the same amount of the total energy consumption predicted in Figure 4, I explored the following two hypothetical scenarios for 2030 (Table 3).

**Table 3.** Hypothetical Scenarios of Energy Mixes in 2030.

| Scenario | Nuclear energy | Wind | Hydro-power | Solar | Biomass | Natural Gas | Oil | Coal |
|---|---|---|---|---|---|---|---|---|
| **A** | 10% | 10% | 10% | 10% | 10% | 20% | 15% | 15% |
| **B** | 20% | 15% | 15% | 15% | 5% | 25% | 5% | 0% |

In the scenario A, we switch to 50% renewable and nuclear energy, while in Scenario B, we make a 70% switch to renewable and nuclear energy sources with no coal use. To make the projections for 2030, we recalculated the energy mix index based on Scenarios A and B and then calculated the expected $CO_2$ emissions and reductions (Table 4 and illustrated in Figure 6).

**Table 4.** Predicted $CO_2$ emissions (GtCO2e) in 2030 with different energy mixes.

| Scenario | $CO_2$ emissions (GtCO2e) in 2030 | Reduction in carbon emissions |
|---|---|---|
| Current trend | 39.15 | -- |
| **A** | 26.26 | 33% |
| **B** | 12.96 | 67% |

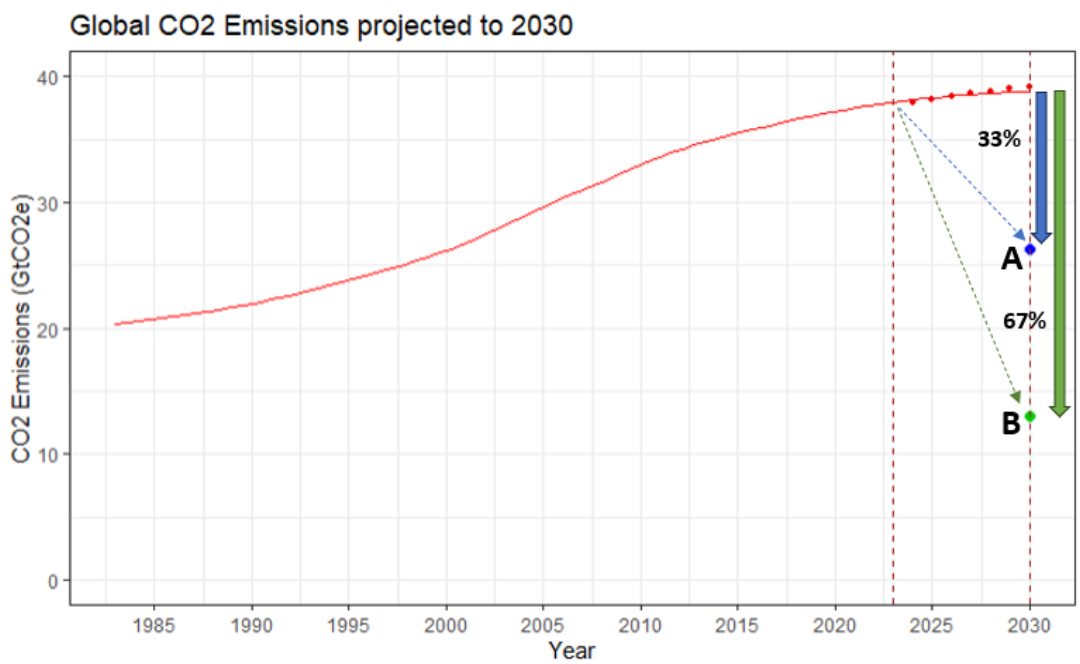


**Figure 6.** The projected $CO_2$ emissions in 2030 under two different energy mixes.

As shown in Table 4 and Figure 6, while both global emissions and energy consumption are projected to rise, increasing the shares of renewable and nuclear energy up to 50% and 70% under Scenarios A and B, respectively, could lead to a reduction in $CO_2$ emissions by 33% and 67% by 2030.

## 4. Discussion and Conclusions

In this study, I developed and applied a flexible model to determine the environmental impact of total $CO_2$ emissions resulting from energy consumption. I considered the impact of various energy mixes from main energy sources and predicted that $CO_2$ emissions will continue to rise from the current state until 2030 at a rapid pace. Using the developed model, I explored two scenarios of increasing the portions of renewable and nuclear energy up to 50% and 70%, which lead to reductions in $CO_2$ emissions by one-third and two-thirds by 2030, respectively.

This model is general to accommodate various changes in the proportions of the eight primary energy sources, which could be useful for policymakers to explore the patterns of change and impacts on $CO_2$ emissions when using different energy sources and mixes. In practice, it may be challenging to make a significant switch due to the social-economic status of countries or regions or technical difficulties. For example, when solar energy is inadequate during the night or on raining days, stored renewable energy may be used.

There are several limitations in my study. First, the model was developed using limited data from online public data sources over the last 40 years, which were not sufficient to train a more complex model. Second, the global energy consumption and energy mix data were used without considering the large country-to-country variabilities. For instance, the US uses more natural gas for primary energy, while other countries may use more hydro and wind power. Additionally, to make prediction of the carbon emissions in 2030, I assumed that both the use of energy source and the pattern of power consumption would follow the similar trends as the last four decades, despite adding two change-points to account for the rate changes over the years.

Our findings have important implications. With the recent technological advances and the global expansion of use in AI and HPC, there is an unprecedented demand for power usage, which is predicted to negatively impact greenhouse gas emissions and global warming. However, energy sources are not made equal – I have presented data that using more renewable and nuclear energy will significantly reduce the $CO_2$ emissions. When switching to renewable and nuclear energy, attention should be paid not long to their impact on $CO_2$ emissions, but also to water consumption, waste, and other environmental impacts.

Many large companies and hyperscale data centers have already started moving towards the target of “zero carbon” to reduce their carbon footprint (Manganelli et al., 2021; Silva et al.;

Singla, 2023). Sustainable and "green" strategies should be adopted to reduce the environmental impact of $CO_2$ emissions (Quaschning, 2019; Rojey, 2009). First, various industries should start to use or switch to more the renewable power sources with low carbon emissions, such as wind, solar, and hydropower. In addition, industries should improve their system efficiency and enhance the utilization rate for the same amount of energy, such as optimizing their energy use management with AI. Finally, governments and agencies worldwide and within the United Nations should enforce more regulations on $CO_2$ emissions and standard greenhouse gas protocols (International Energy Agency, 2024), ensuring companies in different sectors to align their practices with the environmental standards.

In conclusion, this study demonstrates that switching to more renewable and nuclear energy will reduce the $CO_2$ emissions significantly. Enhanced global control, regulations, and environmental standards would minimize the environmental impact of $CO_2$ emissions and better protect our environment for the future.

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